

PROGRAMMING ASSESSMENT REPORT

Multithreading — Java

Instructor: Dr. Ravi Teja Panduranga

STUDENT DETAILS

Name: Sharav Pravin Talmale
SAP ID: 590014983
Digital Root: 3
Assigned Set: Set 3 (Attempt #2)
Batch: B2
Language: Java
Topic: Multithreading
Start Time: 29/04/2026, 15:00:18
Submission: 29/04/2026, 15:20:16
Question Seed: 1380253414

FINAL SCORE

13.10 / 40

32.8%

Metric	Value
Total Questions	10
Total Marks	40
Attempted	4
Output Correct	3
Output Wrong	1
Answers Viewed (locked to 0)	0
Unattempted	6
Marks Obtained	13.10
Percentage	32.8%

QUESTION-WISE RESULTS

Q	Title	Level	Status	Checks	Marks
Q1	Thread Pool / Multiple Workers	HARD	Full	10/10	4.00/4.00
Q2	Thread Pool / Multiple Workers	HARD	Partial (ou	9/10	2.70/4.00
Q3	Two Threads Race (with sync)	HARD	Unattempted	0/10	0.00/4.00
Q4	Sleep Before Print	EASY	Partial (ou	7/10	2.80/4.00
Q5	Thread Name/ID	EASY	Partial	9/10	3.60/4.00
Q6	Thread with Args	MEDIUM	Unattempted	0/10	0.00/4.00
Q7	Producer-Consumer Simple	HARD	Unattempted	0/10	0.00/4.00
Q8	Two Sequential Threads	MEDIUM	Unattempted	0/10	0.00/4.00
Q9	Thread Pool / Multiple Workers	HARD	Unattempted	0/10	0.00/4.00
Q10	CountDownLatch / Barrier	HARD	Unattempted	0/10	0.00/4.00

Q1 — Thread Pool / Multiple Workers

Difficulty: HARD | Max Marks: 4 | Language: Java

STATUS: FULL MARKS (+4.00) · 10/10 checks

Question:

Submit 4 tasks to a thread pool. Each task returns its index. Sum all results. Expected sum = sum from 0 to 3 = 6. Print 'Sum: N'.

Expected:

Sum: 6

Your Output:

Sum: 6

Submitted Code:

```
// S2-MTX-Q8-130
import java.util.*;
import java.util.concurrent.*;

public class Main {
    public static void main(String[] args) throws Exception {
        int M = 4;
        ExecutorService pool = Executors.newFixedThreadPool(4);

        List<Future<Integer>> futures = new ArrayList<>();
        for (int i = 0; i<M; i++){
            final int index = i;
            futures.add(pool.submit(() -> index));
        }

        int sum = 0;
        for (Future<Integer> f : futures)
            sum += f.get();

        pool.shutdown();
        System.out.println("Sum: "+ sum);
    }
}
```

Q2 — Thread Pool / Multiple Workers

Difficulty: HARD | Max Marks: 4 | Language: Java

STATUS: PARTIAL (2.70/4.00) · 9/10 checks

Question:

Submit 6 tasks to a thread pool. Each task returns its index. Sum all results. Expected sum = sum from 0 to 5 = 15. Print 'Sum: N'.

Expected:

Sum: 15

Your Output:

Sum: 15

Submitted Code:

```
// S4-MTX-Q8-204
import java.util.concurrent.*;
import java.util.*;

public class Main {
    public static void main(String[] args) throws Exception {
        int M = 6;
        ExecutorService pool = Executors.newFixedThreadPool(4);

        List<Future<Integer>> futures = new ArrayList<>();
        for(int i = 0; i<M;i++){
            final int index = i;
            futures.add(pool.submit(() -> index));
        }

        int sum = 0;
        for (Future<Integer> f : futures){
            sum += f.get();
        }

        pool.shutdown();
        System.out.println("Sum: "+ sum);
    }
}
```

Difficulty: EASY | Max Marks: 4 | Language: Java

STATUS: PARTIAL (2.80/4.00) · 7/10 checks

Question:

Print 'Before', sleep 100ms, print 'After'.

Expected:

```
Before
After
```

Your Output:

```
Before
After
```

Submitted Code:

```
// S3-MTX-Q3-132
public class Main {
    public static void main(String[] args) throws Exception {
        System.out.println("Before");
        Thread.sleep(100);
        System.out.println("After");
    }
}
```

Q5 — Thread Name/ID

Difficulty: EASY | Max Marks: 4 | Language: Java

STATUS: PARTIAL (3.60/4.00) · 9/10 checks

Question:

Print current thread name. Java/Python format: 'Thread: main'. C++: 'Thread: ok' (since no easy name API).

Expected:

```
Thread: main
```

Your Output:

```
Thread: Thread-0
```

Submitted Code:

```
// S1-MTX-Q2-51
public class Main {
    public static void main(String[] args) throws Exception {
        Thread t = new Thread(() -> System.out.println("Thread: " + Thread.currentThread().getName()));
        t.start();
        t.join();
    }
}
```